

Intra-Inter Graph Representation Learning for Protein-Protein Binding Sites Prediction

Wenting Zhao , Gongping Xu, Long Wang, Zhen Cui , Tong Zhang , and Jian Yang 

Abstract—Graph neural networks have drawn increasing attention and achieved remarkable progress recently due to their potential applications for a large amount of irregular data. It is a natural way to represent protein as a graph. In this work, we focus on protein-protein binding sites prediction between the ligand and receptor proteins. Previous work just simply adopts graph convolution to learn residue representations of ligand and receptor proteins, then concatenates them and feeds the concatenated representation into a fully connected layer to make predictions, losing much of the information contained in complexes and failing to obtain an optimal prediction. In this paper, we present Intra-Inter Graph Representation Learning for protein-protein binding sites prediction (IIGRL). Specifically, for intra-graph learning, we maximize the mutual information between local node representation and global graph summary to encourage node representation to embody the global information of protein graph. Then we explore fusing two separate ligand and receptor graphs as a whole graph and learning affinities between their residues/nodes to propagate information to each other, which could effectively capture inter-protein information and further enhance the discrimination of residue pairs. Extensive experiments on multiple benchmarks demonstrate that the proposed IIGRL model outperforms state-of-the-art methods.

Index Terms—Graph neural network, inter-protein information propagation, intra-protein information enhancement, mutual information maximization, protein-protein binding sites prediction.

I. INTRODUCTION

PROTEIN is the basic substance of all life and is essential for maintaining various functions of the body. Proteins are usually composed of a sequence of amino acids that fold three-dimensional structures by interacting with other residues to carry out necessary functions. In this paper, we focus on protein-protein binding sites prediction between ligand and receptor proteins as shown in Fig. 1, which is crucial in understanding the role of proteins and helpful for drug discovery

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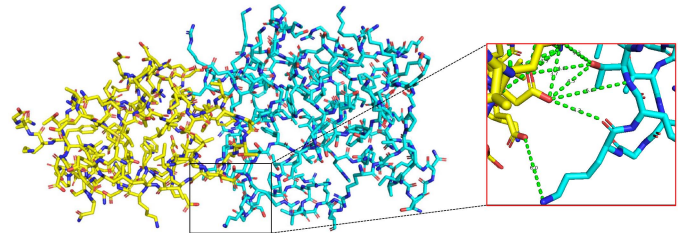


Fig. 1. Stick view of the protein complex named 7CEI, bound from a ligand protein (cyan) and a receptor protein (yellow). The zoomed-up area in the red box shows detailed interactions (green dotted lines) of amino acids between glutamic acid in the receptor and lysine in the ligand.

and design [1], [2], [3]. Protein-protein binding sites prediction needs to classify all residue pairs between ligand and receptor proteins, which is a challenging task [2], [4], [5]. There are many biological experimental methods such as nuclear magnetic resonance (NMR) [6], X-ray crystallography [7] and high-throughput screening [8] used to measure the positions of binding sites. However, the experiment-based methods are laborious and expensive. In addition, traditional machine learning methods also require hand-crafted features that need to be carefully selected by domain experts, usually including surface accessibility, sequence conservation, residue properties such as hydrophobicity and charge, and various shape descriptors [1], [9].

It is a natural way to represent protein as a graph. With the development of deep learning on object detection, semantic segmentation, natural language processing, etc., GCN [10] was first proposed by Kipf and Welling in 2016, since then graph neural networks have received great attention and made progress in many learning tasks such as node classification [10], [11], [12], [13], [14], [15], [16], [17], graph classification [18], [19], [20], [21], [22], [23], [24], and link prediction [25], [26], [27], [28], [29]. Graph neural network breaks through the limitation of regular grids such as images, and extends the application of convolution operators to a wider range of data that can be represented by graphs, where nodes can represent any kind of entity and edges indicate some relationship between entities. Message passing neural network [30] generalized several graph convolutional networks into a unified framework, and defined graph convolution as three phases, i.e., first propagate messages from neighbors, then aggregate the neighbor messages by designing favorable aggregator, finally update central node by fusing the aggregated and itself messages. Recently graph neural networks have been widely extended to various domains such as

social networks [31], [32], recommendation systems [33], [34], point cloud [35], [36], and biochemical [37], [38], etc.

Fout et al. [1] first enabled graph neural network to predict protein interaction interfaces/binding sites by building proteins as graphs with residues as nodes and closer residues as edges, where graph neural network is used to convolve on ligand and receptor protein graph separately, then the output of node representations of ligand and receptor protein graphs are concatenated and fed into a fully connected layer to predict whether an interaction exists for node pair. However, the vanilla graph neural network usually learns suboptimal node representation due to the plain average/sum aggregation over neighbor nodes, losing the structural and semantic information. Besides, although protein-protein binding sites prediction is a local problem between residues, it is helpful to encode the global information into the node representation. To this end, increasing network depth is an intuitive way to incorporate global node representation, but it usually leads to over-smoothing and weakens the discrimination of nodes. Another limitation is that simply concatenating the individual node representation of ligand and receptor protein graphs as input to the classifier could not adequately capture the pairwise representation across proteins/graphs.

To remedy these deficiencies, we present intra-inter graph representation learning (IIGRL) to enrich residue/node representation. For intra-graph representation learning, we propose to maximize the mutual information between the local node representation and global graph summary to encourage node representation to embody the global information of protein. The mutual dependence can help the learned node representation obtain more discriminative information and reveal the potential relations among global nodes. For inter-graph representation learning, we construct the separate ligand and receptor protein graphs as a whole graph and learn affinity between any two nodes of ligand and receptor protein, then perform the inter-graph convolution with the affinity score, which could propagate the node representation between ligand protein and receptor protein to each other. As a result, residues from receptor protein not only possess the information of other nodes in the receptor protein but also carry the semantic information about ligand protein and vice versa. More information encoded could further improve the node representation and enhance the discrimination of node pairs. Summarily, the main contributions of this paper are as follows:

- We present Intra-Inter Graph Representation Learning for protein-protein binding sites prediction (IIGRL). We propose to maximize the mutual information between the local node representation and global graph summary to encourage node representation to embody the global information in the protein graph.
- We explore constructing the separate ligand and receptor graphs as a whole graph and learn affinities between residues across protein so that the learned node representation can be propagated to each other and enhance the discrimination of node pairs.
- We perform extensive experiments on multiple benchmarks and demonstrate that the proposed IIGRL consistently outperforms baselines.

II. RELATED WORK

Here, we briefly introduce the related work about graph neural networks, mutual information estimation and protein-protein interaction.

Graph Neural Networks: Kipf and Welling presented a basic and effective graph convolution form [10], which is widely adopted to learn node representation. Since then, plenty of GNN variants [11], [12], [13], [14], [22], [23], [24] are proposed, where GAT [12] learns different weights for different neighbors to highlight the importance of more similar neighbor. Message passing neural network [30] generalizes several graph convolutional networks into a unified framework. GraphSAGE [11] learns representation by sampling and aggregating node features in a local neighborhood. JKNet [13] introduces the residual connection into the graph neural network and aggregates features of different layers by max-pooling, concatenation, or LSTM-attention. GIN [22] presents a theoretical framework to analyze the expressive power of GNNs and proves that sum aggregation is a superior way for graph classification task. DGCN [39] incorporates both local and global structural information into a two-stream learning framework. Summarily, convolution operation on graphs is a weighted-sum function over the neighbors, which characterizes the impact exerted on the target node from its neighboring ones. More detailed survey can be found in [40].

Mutual Information Estimation: Mutual information (MI) captures non-linear statistical dependencies between variables and thus could act as a measure of true dependence [41]. Since it's challenging to estimate the mutual information accurately between high dimensional continuous random variables, MINE [42] uses a lower bound to the MI based on the Donsker-Varadhan representation [43] of the KL-divergence. Then DIM [44] adopts JSD [45] and infoNCE [46] as an alternative to maximize the mutual information. DGI [47] extends DIM to the graph domain by maximizing MI between local-global pairs. Since then, increasing attention [17], [19], [48], [49] is paid to extending DGI. HDMI [48] extends DGI to three variables and is applied to multiplex networks. N2N [17] is designed to maximize the mutual information between nodes and neighborhoods. InfoGraph [49] and SUGAR [19] apply mutual information constraint on graph classification task.

Protein-Protein Interaction: Graph neural networks show a promising tackle to encode relationships of various entities. Recently research community gradually introduces GNNs into the field of biological sciences [1], [2], [37], [50], [51], [52], [53], [54]. Fout et al. [1] enabled graph neural network for protein interaction interface/binding sites prediction, where residues as nodes and closer residues as edges, graph neural network is used to convolve on ligand and receptor proteins separately, the output of node representation of ligand and receptor protein graphs are concatenated to predict whether there is an interaction for the node pairs. Deep Tensor Neural Networks (DTNN) [52] employs element-wise multiplication to combine the node features and corresponding edge features, where edges serve as gates to help control information from the corresponding nodes. Then the combined features in the neighborhood are aggregated by summing operation and fed into fully connected layer. Siamese

So the total loss is something like:
 $L = L_CS + \lambda * L_MI$

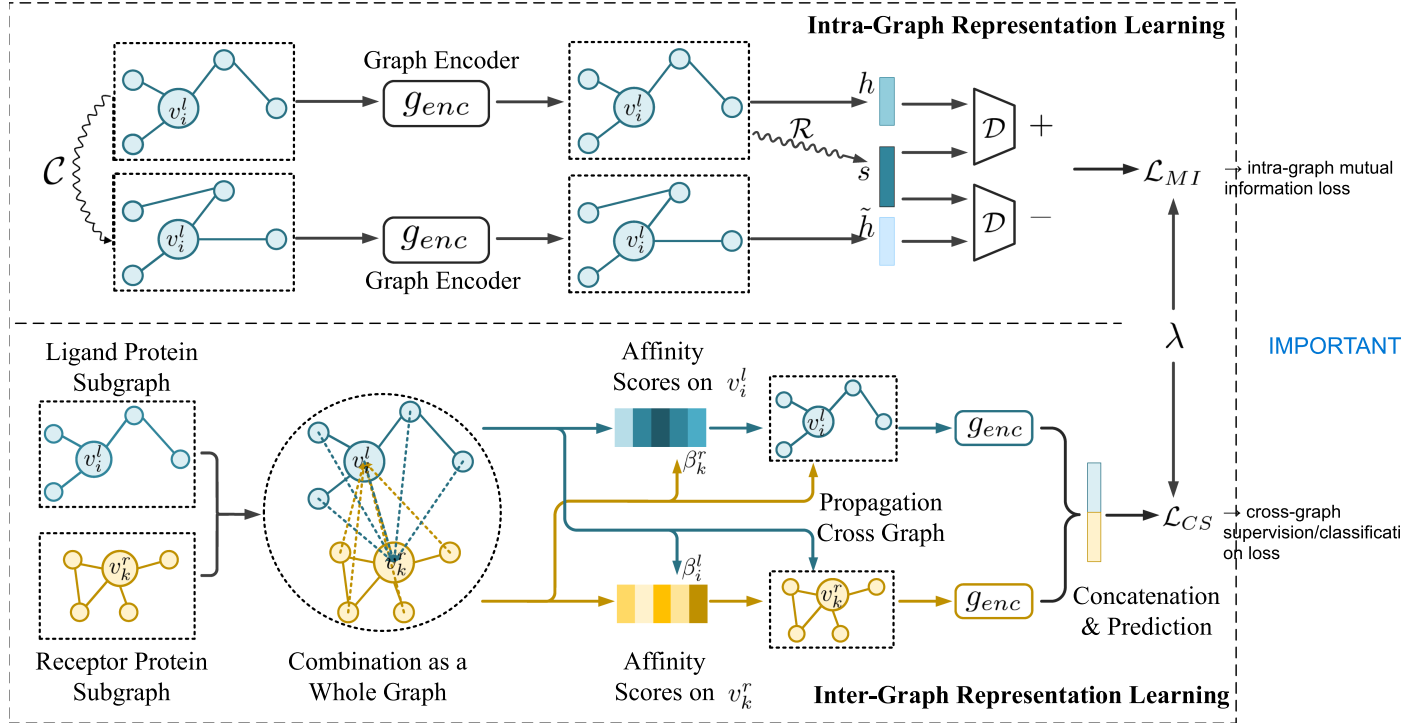


Fig. 2. The framework of our proposed Intra-Inter Graph Representation Learning (IIGRL) for protein-protein binding sites prediction. The upper is intra-graph representation learning, which is designed to improve node/residue representation within protein graph. The bottom is inter-graph representation learning, exploring how to propagate information across proteins and further enrich node representation.

Atomic Surfacelet Network (SASNet) [55] voxelizes all amino acids into 4D grids. The first three dimensions are the spatial information and the last dimension is the one-hot encoding of the atom's element type. The voxelized representations are then fed into a 3D siamese-like CNN, followed by concatenation operation and dense layers to classify whether two amino acids interact or not. Liu et al. [2] formulated the protein interface prediction as a 2D dense prediction problem. Graph convolution is used to aggregate structural information, and sequential modeling (SM) is used to restore the sequential information from original amino acid chains. Then the high-order pairwise interaction (HOPI) is performed for any two amino acids from different proteins. Finally, 2D CNN layers are used for dense prediction. Pittala et al. [51] combined the advantages of graph convolution and attention mechanism in an integrated model for predicting both epitopes and paratopes, where the former aggregate properties across local regions in a protein and the latter explicitly encode the context.

Different from the above methods, we maximize mutual information to enhance the intra-protein representation learned by GNN while we construct two proteins as a whole graph to further propagate the inter-protein information to benefit the protein-protein binding sites prediction.

III. PRELIMINARY

In this section, we first construct the ligand and receptor proteins as graphs and give notations description, then formulate the task of protein-protein binding sites prediction.

Graph Construction: For ligand and receptor protein, each of which consists of a sequence of residues folding certain 3D structures to play specific functions. Given the 3D coordinates of each atom in the residue, we can compute the Euclidean distance between any two atoms of two residues, and we regard the average distance of atoms as the distance between corresponding residues, which is then passed through a Gaussian function:

$$\hat{d}_{ij} = e^{-\frac{d_{ij}^2}{\sigma^2}},$$

Nodes = residues, Edges = connections between Top-k nearest residues, Edge weights = Gaussian similarity scores

where d_{ij} is the raw distance between residue i and j , σ is a hyperparameter used to scale the raw distance.

After obtaining the distance, we select the Top-k closer residues as direct neighbors and give a link, thus

$$\mathcal{N}_i = \{j | \text{Top}_k(\hat{d}_{ij}), j = 1, 2, \dots\}.$$

As a result, protein is constructed as the graph in which residues serve as nodes, and edges are defined above.

Notations Description: Define ligand protein graph as $\mathcal{G}^l(\mathcal{V}^l, \mathcal{E}^l)$, where $\mathcal{V}^l$ represents nodes with the number $|\mathcal{V}^l| = n_l$, $\mathcal{E}^l$ is the set of edges with the number $|\mathcal{E}^l| = m_l$. Besides, each node has a d_v -dimensional feature $\mathbf{x} \in \mathbb{R}^{d_v}$ depicting some natures of residue. Each edge is also associated with a d_e -dimensional feature vector $\mathbf{e} \in \mathbb{R}^{d_e}$ indicating affinity between amino acids. Stacking all nodes and edges feature vector as $\mathbf{X}^l \in \mathbb{R}^{n_l \times d_v}$ and $\mathbf{E}^l \in \mathbb{R}^{m_l \times d_e}$.

Samely, receptor protein graph is defined as $\mathcal{G}^r(\mathcal{V}^r, \mathcal{E}^r)$, where $\mathcal{V}^r$ represents nodes with the number $|\mathcal{V}^r| = n_r$, $\mathcal{E}^r$ is the set of edges with the number $|\mathcal{E}^r| = m_r$. Both node

and edge have a feature description, denoted as $\mathbf{x} \in \mathbb{R}^{d_v}$ and $\mathbf{e} \in \mathbb{R}^{d_e}$, separately. Stacking all nodes and edges feature vector as $\mathbf{X}^r \in \mathbb{R}^{n_r \times d_v}$ and $\mathbf{E}^r \in \mathbb{R}^{m_r \times d_e}$.

Problem Formulation: Protein-protein binding sites prediction aims to predict whether there is an interaction between any two residues from different proteins. Following work [1], [2], we approach this as a binary classification problem. As above, we denote both ligand and receptor proteins as graphs, there are n_l and n_r nodes in the ligand and receptor graph respectively, so we need to carry out $n_l \times n_r$ predictions to classify all residue pairs. If there exists interaction, the label is set to 1, otherwise 0. In this paper, our goal is to develop a mapping $f(\mathcal{G}^l(\mathbf{X}^l, \mathbf{E}^l), \mathcal{G}^r(\mathbf{X}^r, \mathbf{E}^r)) \rightarrow y : \{0, 1\}$ that could encode both the node and edge features extracted from protein graph and learn a better node/residue representation by capturing intra-graph global information as well as propagating cross-graph information.

IV. METHODOLOGY

In this section, we propose Intra-Inter Graph Representation Learning (IIGRL) for protein-protein binding sites prediction, which jointly learns local and global intra-graph node information as well as propagates cross-graph semantic information. We first describe the overall framework of IIGRL, then explain in detail how node representation is improved within a single protein graph and how semantic information between ligand and receptor protein graphs are propagated to each other.

A. Overview

Considering the irregularity of protein structure, graphs are constructed to characterize them by taking residues as nodes and defining spatial relationships as edges. The architecture of our proposed Intra-Inter Graph Representation Learning (IIGRL) for protein-protein binding sites prediction is shown in Fig. 2, which is composed of two learning stages. The upper one is intra-graph representation learning, consisting of two main learning processes, i.e., graph encoding and mutual information maximization. Graph encoding explicitly aggregates the features of local neighbors into the central node, while mutual information maximization between local node representation and global graph summary could promote the learned node representation to contain global information of graph. The bottom one is inter-graph representation learning, combining ligand protein and receptor protein as a whole graph and propagating their information to each other. Taking one node/residue v_i^l in ligand protein graph as an example, inter-graph learning would first give edges to node v_i^l from all nodes of receptor protein graph, then compute affinity scores and propagate information of receptor protein to node v_i^l . Also, the information of ligand protein could be propagated to node v_k^r in receptor protein graph. Finally, the learned residue representations from ligand and receptor proteins are combined and fed into a classifier to predict the binding sites. The objective of mutual information maximization and prediction loss function are jointly optimized by backpropagation. In the following sections, we describe each module in detail.

B. Intra-Graph Representation Learning

For ligand protein graph $\mathcal{G}^l(\mathbf{X}^l, \mathbf{E}^l)$ and receptor protein graph $\mathcal{G}^r(\mathbf{X}^r, \mathbf{E}^r)$, we employ two graph encoders $g_{enc}^l : \mathbb{R}^{n_l \times d_v} \times \mathbb{R}^{m_l \times d_e} \rightarrow \mathbb{R}^{n_l \times d}$ and $g_{enc}^r : \mathbb{R}^{n_r \times d_v} \times \mathbb{R}^{m_r \times d_e} \rightarrow \mathbb{R}^{n_r \times d}$ as work [2] to learn node representations. For node v_i^l in ligand graph, we first incorporate its neighbor node features and corresponding edge features as,

$$\mathbf{z}_j^l = f_v^l(\mathbf{x}_j^l) + f_e^l(\mathbf{e}_{ij}^l), \quad (1)$$

where $\mathbf{x}_j^l$ is the feature vector of node $v_j^l \in \mathcal{N}_i^l$, $\mathbf{e}_{ij}^l$ is the edge feature that links node v_i^l and v_j^l . f_v^l and f_e^l are two functions imposed on the node and edge features, separately. In practice, we use linear mapping for f_v^l and f_e^l .

Considering that different neighbors have different features and different distances to the central node v_i^l , attention scores are computed to highlight more informative neighbors. Formally, the weighted aggregation process can be written as,

$$\mathbf{h}_i^l = \text{Relu}(f_c^l(\mathbf{x}_i^l) + \sum_j \alpha_j \mathbf{z}_j^l), \quad \text{After message passing x becomes h}$$

$$\alpha_j = \text{softmax}_j(\mathbf{z}_j^l \mathbf{W}^l), \quad v_j^l \in \mathcal{N}_i^l, \quad (2)$$

Through this stage, important neighbors get larger weights α_j where $\mathcal{N}_i^l$ is the set consisting of neighbors of node v_i^l . $\mathbf{W}^l$ is trainable parameter matrix to learn the attention score. Softmax function is employed to normalize the attention score. f_c^l is a linear transformation to adaptively adjust the dimension of node features and fuse its neighbors' features. Relu is the activation unit.

However, graph convolution usually encodes local neighborhood information, it is not easy to capture global information. In order to improve node representations, we propose to maximize the mutual information between the local node representation and global graph summary to encourage node representation to embody the global information of protein graph. Specifically, we compute the graph-level representation by summarizing node representations through a readout function $\mathcal{R} : \mathbb{R}^{n_l \times d} \rightarrow \mathbb{R}^d$,

$$\mathbf{s}^l = \mathcal{R}(\mathbf{H}^l) = \sigma\left(\frac{1}{n_l} \sum_i \mathbf{h}_i^l\right), \quad (3)$$

where $\mathbf{H}^l \in \mathbb{R}^{n_l \times d}$ is the representation of all nodes in the ligand protein graph, $\mathcal{R}$ is the readout function, which could be sum, mean, or max, etc., here we employ the mean function. σ is the logistic sigmoid nonlinearity.

The mutual information between the node representation $\mathbf{h}_i^l$ and global graph summary $\mathbf{s}^l$ is defined as

$$I_{\Theta}(\mathbf{h}_i^l; \mathbf{s}^l) = \int d\mathbf{h}_i^l d\mathbf{s}^l p(\mathbf{h}_i^l, \mathbf{s}^l) \log\left(\frac{p(\mathbf{h}_i^l, \mathbf{s}^l)}{p(\mathbf{h}_i^l)p(\mathbf{s}^l)}\right), \quad (4)$$

where Θ is network parameters. It is difficult to exactly compute the mutual information due to the high-dimensional representation vector. In this work, we adopt the Jensen-Shannon divergence to approximately maximize the mutual information as,

$$\mathcal{L}_{MI} = -\mathbb{E}[\log \mathcal{D}(\mathbf{h}_i^l; \mathbf{s}^l)] - \mathbb{E}[\log(1 - \mathcal{D}(\tilde{\mathbf{h}}_i^l; \mathbf{s}^l))], \quad (5)$$

watch the GAN video in DL playlist for this formula's derivation

The discriminator uses JSD to maximize mutual information by distinguishing:

* real pairs from joint distribution $p(\mathbf{h}, \mathbf{s})$

* fake independent pairs from $p(\mathbf{h})p(\mathbf{s})$ authorized licensed use limited to the terms of the applicable license agreement with IEEE. Restrictions apply.

Just like GANs distinguish real vs fake data distributions.

Same as work [47], we give a corruption $\mathcal{C}$ to the protein graph, then generate corrupted graph $\tilde{\mathcal{G}}^l(\tilde{\mathbf{X}}^l, \tilde{\mathbf{E}}^l) = \mathcal{C}\{\mathcal{G}^l(\mathbf{X}^l, \mathbf{E}^l)\}$ as input of the same graph encoder, and $\mathbf{h}_i^l$ is the node representation learned from the corrupted graph $\tilde{\mathcal{G}}^l$. $\mathcal{D}$ is a bilinear discriminator function.

This objective effectively maximizes the mutual information between local node representation $\mathbf{h}_i^l$ and global summary $\mathbf{s}^l$ based on the Jensen Shannon divergence between the joint and the product of marginals. The mutual dependence can help the learned node representation obtain more discriminative information and thus reveal potential relations among global nodes. Note that the same operation is performed on the receptor protein graph.

C. Inter-Graph Representation Learning

Previous work just simply concatenates the individual node representation of ligand and receptor protein graphs learned by graph convolution as the input to the classifier to obtain the final prediction, which can not adequately capture the pairwise representation across graphs. Hence, we explore inter-graph representation learning to fuse respective information in advance.

Specifically, we construct the ligand and receptor protein graphs as a whole graph, for each node in the ligand protein graph, give a link to all nodes in the receptor protein graph, and vice versa. And we introduce two inter-graph/protein encoders $g_{enc}^{l \rightarrow r}$ and $g_{enc}^{r \rightarrow l}$ and perform the inter-graph convolution incorporating inter-protein information to improve the node representation. We compute affinities to assign different importance to the neighbors across graphs. Considering that node representations of ligand and receptor proteins are not in the same feature space, we employ a bilinear transformation to calculate the node affinity between ligand and receptor proteins, which can be expressed as,

$$a_{ik} = (\mathbf{h}_i^l)^\top \mathbf{W} \mathbf{h}_k^r, \quad (6)$$

where $(\mathbf{h}_i^l)^\top$ is the transpose of representation $\mathbf{h}_i^l$ of node v_i in ligand protein, and $\mathbf{h}_k^r$ is the representation of node v_k in receptor protein. $\mathbf{W}$ is learnable parameters of the network.

In order to propagate information from receptor protein to ligand protein, we normalize the affinity score by a softmax function,

$$\beta_k^r = \text{softmax}_k(a_{ik}), \quad k \in \mathcal{V}^r. \quad (7)$$

Next, we aggregate the node representation of receptor protein for node v_i^l in ligand protein as,

$$\mathbf{h}_i^{r \rightarrow l} = \sum_k \beta_k^r \mathbf{h}_k^r, \quad k \in \mathcal{V}^r. \quad (8)$$

Then we fuse the representation aggregated from receptor protein with the representation of node v_i^l to update the node v_i^l in the ligand protein,

$$\bar{\mathbf{h}}_i^l = f_a^l(\mathbf{h}_i^l || \mathbf{h}_i^{r \rightarrow l}), \quad (9)$$

where $||$ represents concatenation, f_a^l is a nonlinear transformation used to reduce the dimension of the concatenation.

Similarly, we propagate and aggregate information from ligand and protein to update the representation of node v_k^r in the receptor protein as follows,

$$\beta_i^l = \text{softmax}_i(a_{ik}^T), \quad i \in \mathcal{V}^l,$$

$$\mathbf{h}_k^{l \rightarrow r} = \sum_i \beta_i^l \mathbf{h}_i^l, \quad i \in \mathcal{V}^l,$$

$$\bar{\mathbf{h}}_k^r = f_a^r(\mathbf{h}_k^r || \mathbf{h}_k^{l \rightarrow r}). \quad (10)$$

After fusing inter-protein information, the same graph encoding layer as described in Section IV-B is employed to extract neighbor information in the respective proteins. Intuitively, this operation can highlight the information about the protein it is in. The output for node v_i^l in ligand protein and node v_k^r in receptor protein can be formulated as,

$$\begin{aligned} \bar{\mathbf{h}}_i^l &= g_{enc}^l(\bar{\mathbf{H}}_{\mathcal{N}_i}^l, \mathbf{E}_{\mathcal{N}_i}^l), \\ \bar{\mathbf{h}}_k^r &= g_{enc}^r(\bar{\mathbf{H}}_{\mathcal{N}_k}^r, \mathbf{E}_{\mathcal{N}_k}^r), \end{aligned} \quad (11)$$

where $\bar{\mathbf{H}}_{\mathcal{N}_i}^l$ represents the representations of node v_i^l and its neighbors in ligand protein, and $\mathbf{E}_{\mathcal{N}_i}^l$ is the corresponding edge features.

So far, the residues/nodes possess the semantic and structural information of both ligand and receptor proteins, and more information is encoded into representations, enhancing the discrimination of nodes.

D. Protein-Protein Binding Sites Prediction

We use the concatenation operation to combine the learned node representation of v_i^l from ligand protein and v_k^r from receptor protein. Then a fully connected layer (FC) is adopted as a classifier to output prediction of binding site,

$$\hat{y}_{ik} = \text{FC}(\bar{\mathbf{h}}_i^l || \bar{\mathbf{h}}_k^r). \quad (12)$$

We adopt binary cross-entropy loss as the objective,

$$\mathcal{L}_{CS} = \frac{1}{\mathbb{N}} \sum_{ik} -[y_{ik} \log(\hat{y}_{ik}) + (1 - y_{ik}) \log(1 - \hat{y}_{ik})], \quad (13)$$

where $\mathbb{N}$ is the number of training samples.

Besides, we jointly optimize the cross-entropy loss and Jensen Shannon divergence as shown in Fig. 2, and a hyperparameter λ is used to balance these two terms. Thus the final objective is formulated as,

$$\mathcal{L} = \mathcal{L}_{CS} + \lambda \mathcal{L}_{MI}. \quad (14)$$

V. EXPERIMENTS

In this section, we carry out extensive experiments to assess the effectiveness of our proposed intra-inter graph representation learning for protein-protein binding sites prediction. We first introduce datasets, baselines, and details of the experimental setting, next we show the experimental results and give some analysis, finally we present ablation study and explain the effect of each component. The code is available at <https://github.com/ProteinsIIGRL/IIGRL>.

Algorithm 1: IIGRL for Protein-Protein Binding Sites Prediction.

Input: ligand graph $\mathcal{G}^l(\mathbf{X}^l, \mathbf{E}^l)$, receptor protein graph $\mathcal{G}^r(\mathbf{X}^r, \mathbf{E}^r)$, intra-graph encoder $g_{enc}^{l/r}$, inter-graph encoder $g_{enc}^{r \rightarrow l/l \rightarrow r}$, readout function $\mathcal{R}$, fully-connected mapping FC.

Output: The predicted label of residue pair i and k .

```

1 for residue/node pair  $(v_i^l, v_k^r)$  do
    // Intra-graph representation learning
2    // ligand graph representation learning
3     $\mathbf{h}_i^l = g_{enc}^l(\mathbf{x}_i^l)$ ,  $\mathbf{s}^l = \mathcal{R}(\mathbf{H}^l)$ ;
4    maximize  $I(\mathbf{h}_i^l; \mathbf{s}^l)$ ;
5    // receptor graph representation learning
6     $\mathbf{h}_k^r = g_{enc}^r(\mathbf{x}_k^r)$ ,  $\mathbf{s}^r = \mathcal{R}(\mathbf{H}^r)$ ;
7    maximize  $I(\mathbf{h}_k^r; \mathbf{s}^r)$ ;
    // Inter-graph representation learning
8    // ligand graph representation learning
9     $\bar{\mathbf{h}}_i^l = f_a^l(\mathbf{h}_i^l || \mathbf{h}_i^{l \rightarrow l})$ ,  $\mathbf{h}_i^{l \rightarrow l} = g_{enc}^{r \rightarrow l}(\mathbf{H}^l, \mathbf{H}^r)$ ;
10    $\bar{\mathbf{h}}_i^l = g_{enc}^l(\bar{\mathbf{H}}_{\mathcal{N}_i}^l, \mathbf{E}_{\mathcal{N}_i}^l)$ ;
11   // receptor graph representation learning
12    $\bar{\mathbf{h}}_k^r = f_a^r(\mathbf{h}_k^r || \mathbf{h}_k^{r \rightarrow r})$ ,  $\mathbf{h}_k^{r \rightarrow r} = g_{enc}^{l \rightarrow r}(\mathbf{H}^l, \mathbf{H}^r)$ ;
13    $\bar{\mathbf{h}}_k^r = g_{enc}^r(\bar{\mathbf{H}}_{\mathcal{N}_k}^r, \mathbf{E}_{\mathcal{N}_k}^r)$ ;
    // Protein-Protein Binding Sites Prediction
14    $y_{ik} = \text{FC}(\bar{\mathbf{h}}_i^l || \bar{\mathbf{h}}_k^r)$ ;
15   min  $\mathcal{L}_{CS}(y_{ik}, y_{ik}^*)$ 
16 end
```

A. Datasets

In experiments, we employ four datasets DB3 [56], DB4 [57], DB5 [58] and DB5.5¹ from the **Docking Benchmark Dataset** (DBD), which is a standard benchmark dataset for assessing binding sites prediction methods [58]. Specifically, DB3 [56] includes 124 cases that are classified into 88 rigid-body cases, 19 medium-difficulty cases, and 17 difficult cases, based on the degree of conformational change at the interface upon complex formation. DB4 [57] adds 52 new complexes over the DB3, an increase of 42%, where 33 rigid-body cases, 11 medium-difficulty cases, and 8 difficult cases. 55 new complexes are added to the DB5 dataset [58] over the DB4, and 41 new complexes are added to the DB5.5 dataset over the DB5. Note that the number of complexes used in experiments as shown in Table I does not match the number in the original dataset. All datasets used in our paper are released by ZLAB.¹ We follow their work [1] and download DB5 dataset from the website.² Then we collect DB4 and DB3 datasets from DB5 according to the list.¹

TABLE I
THE STATISTICS FOR ALL DATASETS

Dataset	Train/Test	Complexes	Positive	Negative
DB3	Train	81	7517 (9.1%)	75170 (90.9%)
	Test	38	3448 (0.2%)	1777517 (99.8%)
DB4	Train	119	10965 (9.1%)	109650 (90.9%)
	Test	52	4531 (0.2%)	2303859 (99.8%)
DB5	Train	175	16004 (9.1%)	160040 (90.9%)
	Test	55	4871 (0.1%)	4953446 (99.9%)
DB5.5	Train	230	20875 (9.1%)	208750 (90.9%)
	Test	41	3816 (0.1%)	4238213 (99.9%)

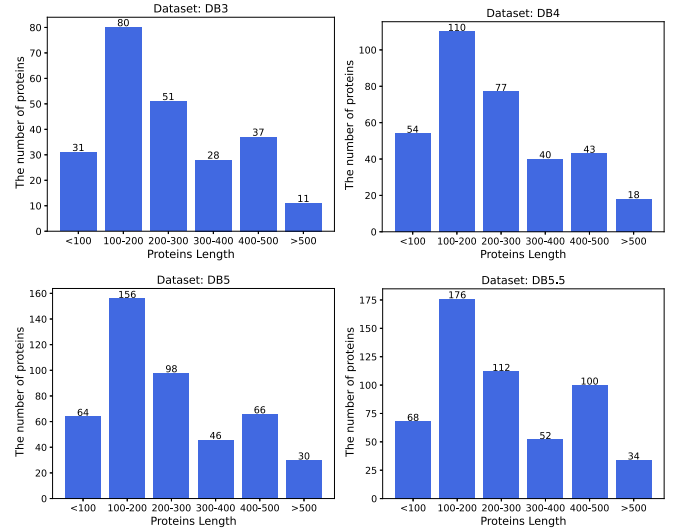


Fig. 3. The length distribution of proteins (each complex consists of two proteins).

The distribution of the number of interacting residues in all datasets is shown in Table I, where interacting residues are Positive pairs, while non-interacting residues are Negative pairs. From Table I, we can find that there are far more Negative pairs than Positive pairs. During training phase, we sample negative samples so that the ratio of positive samples to negative samples is 1:10. In addition, Fig. 3 shows the length distribution of proteins for all datasets, we can observe that most proteins contain 100-300 amino acids. According to statistics, the minimum and maximum protein lengths for the DB3/DB4 dataset are 29 and 1979, respectively. The minimum and maximum protein lengths for the DB5/DB5.5 dataset are 29 and 2128, respectively.

All proteins are obtained from the Protein Data Bank [59],³ where each protein comprises a sequence of amino acids, and the number of amino acids varies from tens to thousands. The structures of proteins/complexes are determined from x-ray crystallography or nuclear magnetic resonance experiments [60] containing the atomic coordinates of each amino acid residue in the protein. For each complex, DBD provides the bound state of two proteins in the complex as well as the unbound state of each protein. The features are computed from the unbound state, and the labels are derived from the structure of the complex.

¹<https://zlab.umassmed.edu/benchmark/>

²<https://github.com/fouticus/pipgcn/tree/master>

³<https://www.rcsb.org/>

If the distance between any two non-Hydrogen atoms from two residues of different proteins is within 6Å, then these two residues are considered to be interacting.

1) *Node and Edge Features*: Considering the irregularity of protein structure, graphs are constructed to describe them by taking residues as nodes and defining spatial relationships as edges. Both nodes and edges are associated with features computed from the amino acid sequence and structure. For all datasets, we follow the work [1] to extract features and labels.

The node features are computed based on different properties of residues. Generally, the features are divided into two types, i.e., sequence features and structural features. Each residue exhibits different electrochemical, geometric properties, levels of conservation, etc. in alignments against similar proteins, where the identity and conservation are quantified by 20 features that capture the relative frequency of 20 amino acids in alignments to similar proteins. Besides, we extract relative Accessible Surface Area(rASA), residue depth, protrusion index, hydrophobicity and half-sphere amino acid composition as structural features that encode important inherent structural information. The sequence and structural features are combined and concatenated together in a consistent order, resulting in 70-dimensional feature vector. We explain node features in detail in the supplementary material.

The edge feature is a 2-dimensional vector representing distance and angle between two residues, respectively. We compute the Euclidean distance between any two atoms from different residues based on the 3D coordinates of each atom, and the average distance of atoms is regarded as the distance between corresponding residues. The feature is a Radial Basis Function (RBF) of this distance. We calculate the angle between the normal vectors of the amide plane of each residue as the relative orientation of two residues. All node and edge features are normalized to between 0 and 1, except for residue conservation features which are standardized.

2) *Protocol*: For the Docking Benchmark (DB) datasets, the training set is generally the complexes from previous version dataset, and the test set is that updated in the current version. For example, we split DB5.5 into a training set corresponding to DB5 and a test set composed of 41 new complexes over DB5. For the DB5 dataset, we use the same training and test set as all baseline methods [1], [2], [50], [52], [55]. For the DB3, DB4, and DB5.5 datasets, we use the fixed training and test samples shown on the website¹. Following the previous works [1], [2], [55], we evaluate the performance by the median area under the receiver operating characteristic curve (MedAUC) across all complexes in the test set. The experiments are repeated 10 times and we use the average of the ten runs as the final performance. For each method, we report the mean and standard deviation of MedAUC across 10 runs.

B. Baselines

We compare our proposed intra-inter graph representation learning for protein-protein binding sites prediction with several state-of-the-arts, including MFN [50], DTNNs [52], NEA [1], SASNet [55], NeiA+HOPI [2]. The baselines developed

different graph neural architectures for node representation learning to predict the interaction.

- MFN is one of the first methods using graphs to represent molecules in chemistry, starting from the common circular fingerprint as an analogy, and designing a set of graph fingerprints. The main advantage of the resulting graph fingerprint is that it is differentiable and thus can be learned end-to-end.
- Deep Tensor Neural Networks (DTNN) employs element-wise multiplication to combine the node features and corresponding edge features, where edges may serve as gates to help control information from the corresponding nodes. Then it aggregates the combined features in the neighborhood by summing operation, followed by two fully connected layers.
- NEA is Node and Edge Average, first using element-wise summation to combine the node features and corresponding edge features, and then aggregating the combined features in the neighborhood by average operation. Another variant is NA that sums the combined features in the neighborhood. Finally, the node is updated by summing the features of the node itself and aggregated neighbors.
- Siamese Atomic Surfacelet Network (SASNet) voxelizes all amino acids into 4D grids. The first three dimensions are the spatial information and the last dimension is the one-hot encoding of the atom's element type. The voxelized representations are then fed through a 3D siamese-like CNN, followed by concatenation operation and dense layers to classify whether two amino acids interact with each other.
- NeiA+HOPI represents proteins as graphs and employs graph neural networks to learn node features. Graph convolution is used to aggregate structural information, and sequential modeling (SM) is used to restore the sequential information from the original amino acid chains. Then the high-order pairwise interaction (HOPI) is performed to build pairwise interactions for any two amino acids from different proteins via concatenation operation. Finally, 2D CNN layers are used for dense prediction. NeiA simply averages on all neighboring features while NeiWA performs a weighted average on the neighborhood in graph convolution layer.

C. Experimental Settings

For intra-graph learning, we employ siamese-like network to encode ligand and receptor protein, in which each graph encoder consists of two graph convolution layers, producing 512-dimensional node representation. For the inter-graph learning, we conduct one-layer graph convolution to propagate cross-protein information and one-layer graph convolution to highlight the intra-protein information. This process is iterated twice. Finally, the residue representations of ligand and receptor proteins are concatenated and fed into a fully connected layer for prediction. Rectified Linear Unit (ReLU) is leveraged as the activation function and dropout is set to 0.4. We train our model with a stochastic gradient descent optimizer (SGD) for 200 epochs, where the learning rate is 0.1. Since the number of positive pairs is far less than that of negative pairs, we adopt the

TABLE II
COMPARISON WITH STATE-OF-THE-ART-METHODS, MEDAUC IS THE METRIC

Methods	DB3	DB4	DB5	DB5.5
MFN [50]	0.875 (0.019)	0.878 (0.009)	0.871 (0.013)	0.929 (0.012)
DTNNs [52]	0.853 (0.017)	0.870 (0.015)	0.880 (0.007)	0.924 (0.013)
NA [1]	0.895 (0.015)	0.902 (0.005)	0.882 (0.007)	0.931 (0.007)
NEA [1]	0.881 (0.009)	0.898 (0.005)	0.898 (0.005)	0.917 (0.008)
SASNet [55]	0.800 (0.012)	0.802 (0.014)	0.876 (0.037)	0.910 (0.008)
NeiA+HOPI [2]	-	-	0.919 (0.015)	-
NeiWA+HOPI [2]	-	-	0.930 (0.016)	-
IIGRL (Ours)	0.912 (0.002)	0.918 (0.003)	0.938 (0.004)	0.940 (0.002)

weighted cross-entropy loss to train our model in an end-to-end manner. Specifically, the weight for positive pairs is 0.1 while 1 for negative ones.

D. Experimental Results

We compare the performance of our IIGRL model against several of the state-of-the-arts: MFN [50], DTNNs [52], NEA [1], SASNet [55], NeiA+HOPI [2]. Note that all results of baseline methods on the DB5 dataset come from the corresponding paper, and results on DB3, DB4 and DB5.5 are obtained by re-running their released code for fair comparison. Table II has shown results clearly indicating that our IIGRL approach achieves the new state-of-the-art performance and obtains a remarkable improvement on all datasets. Specifically, NA and NEA [1] simply employ graph neural network to convolve on ligand and receptor proteins, then concatenate respective representations to predict whether there is an interaction for node pairs. Compared to them, the proposed IIGRL model brings in intra-graph learning and inter-graph learning to improve node representation, thus IIGRL attains performance of 0.912 on DB3, 0.918 on DB4, 0.938 on DB5 and 0.940 on DB5.5. Compared with NEA [1], our method achieves approximately 4% performance gain on the most widely used DB5 dataset, and 3.1%, 2%, 2.3% improvements on DB3, DB4 and DB5.5 respectively. Recent work NeiA+HOPI [2] reports relatively higher performance among those baselines on the DB5 dataset. However, it conducts additional data augmentation by incorporating in-protein pairwise interactions as positive pairs. Compared with this method, our proposed IIGRL still obtains better performance. Since the code of NeiA+HOPI has not been released, the results on other datasets could not be obtained. SASNet introduces 3D siamese-like CNN to extract representation and uses dense layers to classify whether two amino acids interact with each other. In comparison, our method obviously outperforms it, 11.2%, 11.6%, 6.2%, and 3.0% are raised on four datasets respectively. All results above demonstrate the effectiveness of our proposed IIGRL framework and also indicate the robustness against variation of dataset size. We attribute the performance gains to the proposed mutual information maximization of intra-protein/graph learning as well as the information propagation of inter-protein learning, better capturing and encoding the relationship between residues.

TABLE III
THE ABLATION STUDY ABOUT INTRA/INTER-GRAPH LEARNING ON DB5 DATASET

Methods	MedAUC	Recall	F1-score	MCC
NEA [1]	0.898	-	-	-
IIGRL w/o intra-graph	0.929	0.827	0.01	0.058
IIGRL w/o inter-graph	0.911	0.826	0.012	0.06
IIGRL	0.936	0.841	0.014	0.067

E. Ablation Study

As the proposed IIGRL model has achieved promising performance compared to baselines, it is meaningful to make clear the effect of the modules and parameter settings. For this purpose, we conduct several additional experiments to explore IIGRL framework based on DB5 dataset as follows.

The effect of intra-graph learning and inter-graph learning: For clear comparison, we summarize results in Table III, where NEA [1] is a baseline, which means removing intra-graph and inter-graph learning from the IIGRL model. Compared to NEA [1], the performance of “IIGRL w/o intra-graph” is 0.929, and 3.1% improvement is obtained when only adding inter-graph learning. The performance of “IIGRL w/o inter-graph” is 0.911 and 1.3% is improved when only adding intra-graph learning. When both intra-graph and inter-graph learning are added, the model IIGRL achieves state-of-the-art. These results indicate that the proposed single intra/inter-graph learning module is effective in enhancing residues representations.

In addition, we also noticed that the results of F1-score and MCC are generally low, which is caused by the imbalance of positive and negative samples. The negative samples are far more than the positive samples, which makes the number of samples incorrectly predicted as positive (FP) by the model far greater than the number correctly predicted as positive (TP). Similar results are also shown in Table V and Table VI.

The number of graph convolution layers for intra-graph learning: For a comprehensive comparison, we conduct experiments with multiple numbers of graph convolution layers in the graph encoder of intra-graph learning. As illustrated in Fig. 4, we can observe that the MedAUC of stacking two convolution layers is higher than other cases. The aggregated information of neighbors is not sufficient when only one graph convolution layer is employed. Also, more graph convolution layers lead to performance reduction because of the over-smoothing.

TABLE IV
HYPER-PARAMETER SENSITIVITY OF λ ON DB5 DATASET

Methods	0	0.01	0.05	0.1	0.5	1.0
IIGRL	0.929 (0.003)	0.937 (0.004)	0.938 (0.004)	0.938 (0.005)	0.937 (0.004)	0.937 (0.003)

TABLE V
HYPER-PARAMETER SENSITIVITY OF LEARNING RATE ON DB5 DATASET

Learning Rate	MedAUC	Recall	F1-score	MCC
0.05	0.904	0.886	0.01	0.052
0.1(IIGRL)	0.936	0.841	0.014	0.067
0.2	0.928	0.83	0.011	0.054

TABLE VI
HYPER-PARAMETER SENSITIVITY OF DROPOUT ON DB5 DATASET

Dropout	MedAUC	Recall	F1-score	MCC
0	0.917	0.784	0.014	0.063
0.2	0.93	0.831	0.011	0.055
0.4(IIGRL)	0.936	0.841	0.014	0.067
0.6	0.91	0.888	0.01	0.054

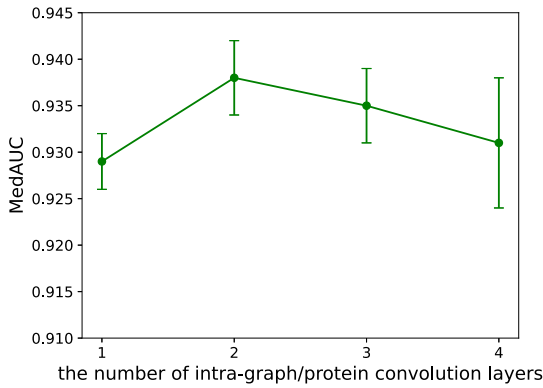


Fig. 4. Comparison on the number of graph convolution layer for intra-graph learning on DB5 dataset.

The number of graph convolution layers for inter-graph learning: To investigate the effect of cross-protein graph convolution in IIGRL, we stack multiple layers and compare the performance between them. The results are shown in Fig. 5, where the performance varies with the number of cross-protein graph convolution layers. When the number of cross-protein graph convolution is set to 0, meaning that the information of ligand and receptor proteins would not be propagated to each other, and the MedAUC is significantly lower. Contrastly, 1.8% performance gain is obtained when one-layer cross-protein graph convolution is adopted, which shows that the information from another protein effectively improves the node/residue representation and promotes the recognition of the residue pair relationships. The best MedAUC is obtained when incorporating two-layer cross-protein graph convolution.

Hyper-parameter sensitivity of λ : In the intra-graph learning module, we maximize the mutual information between the local node representation and global graph summary. We trade off this cost into the final objective function via hyper-parameter λ .

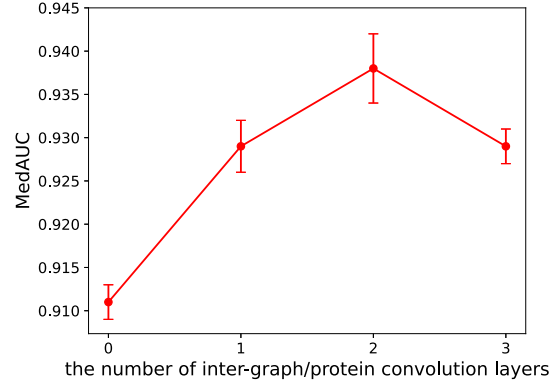


Fig. 5. Comparison on the number of graph convolution layer for inter-graph learning on DB5 dataset.

TABLE VII
TEST TIME (SECOND) ON DB5 DATASET

Methods	average single complex	all complexes
MFN [50]	3.4	184.7
DTNNs [52]	5.2	288.1
NA [1]	5.6	306.9
NEA [1]	7.0	386.5
IIGRL (Ours)	7.3	399.5

Thus we tune the hyper-parameter λ in the range [0, 0.01, 0.05, 0.1, 0.5, 1.0], and corresponding results are shown in Table IV. Overall, we find that the performance is relatively stable with the change of λ . When $\lambda = 0$, that means the term of mutual information dependence is removed, and the performance is obviously lower than others. For too large values of λ , the performance still remains high. This demonstrates that the intra-graph learning module is beneficial to obtain promising results, the mutual information maximization enables the local node/residue to capture global information and be more discriminative.

Hyper-parameter sensitivity of Learning Rate: We set learning rate equal to 0.05, 0.1, and 0.2 to compare their performances. As reported in Table V, when learning rate is 0.1, the performance is best under MedAUC, F1-score and MCC metrics. When the learning rate is 0.05, the best Recall is obtained.

Hyper-parameter sensitivity of Dropout: Specifically, we vary values of dropout in the range [0, 0.2, 0.4, 0.6], and Table VI shows the corresponding results, we can observe that when dropout is equal to 0.4, the performance is better than others under MedAUC and MCC metrics, and comparable for F1-score, while a Recall of 0.888 is obtained when dropout is equal to 0.6.

Time cost: To quantitatively evaluate the time cost of each method, we obtain the test time on DB5 dataset as shown in the Table VII. Compared to NEA [1], the proposed IIGRL

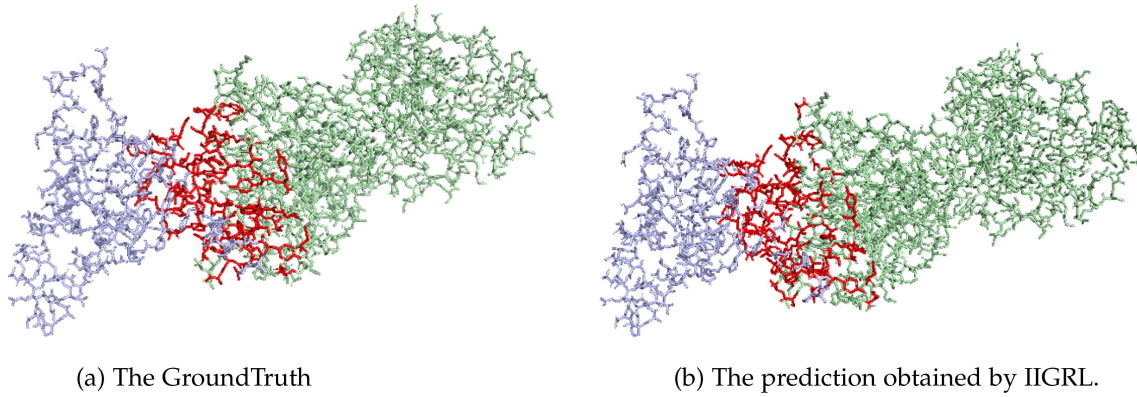


Fig. 6. PyMOL [61] visualizations of test complex (PDB ID 3V6Z). (a) Ligand protein (purple) and receptor protein (green), along with the true interface (red). (b) The red parts denote the learned salient site pairs between the same ligand protein (purple) and receptor protein (green).

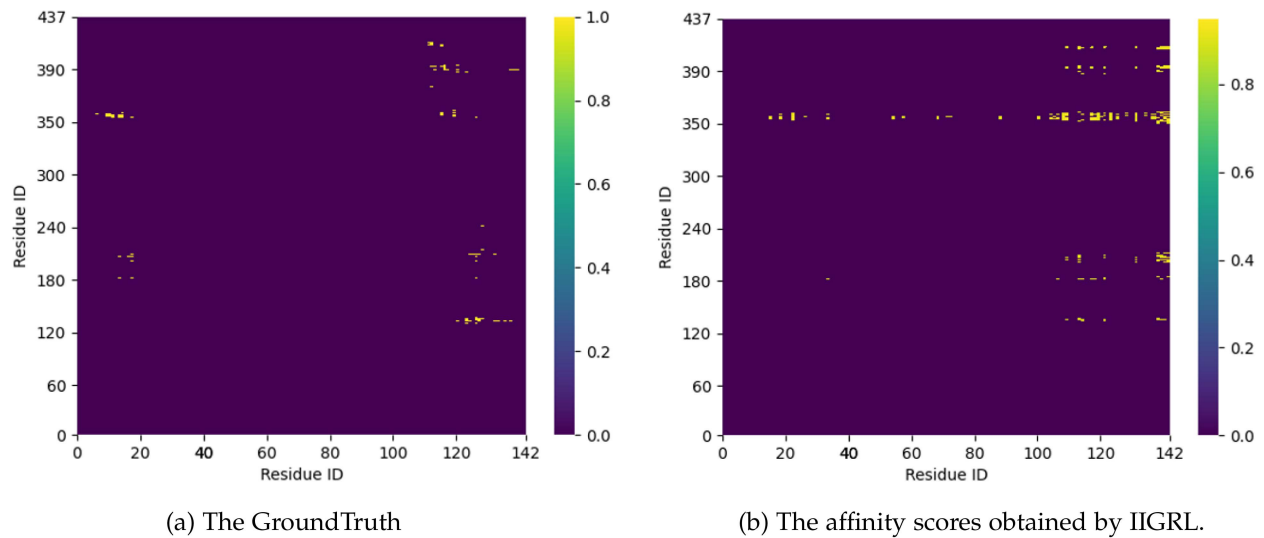


Fig. 7. The heatmap of groundtruth and affinity scores for test complex (PDB ID 3V6Z). The horizontal and the vertical axes represent the residue IDs of the ligand and receptor proteins.

increases intra-graph and inter-graph learning modules, thus time consumption is relatively high.

Visualization of salient pairs of residues: To intuitively understand the effect of our proposed intra-inter graph representation learning for protein-protein binding sites prediction. We visualize the complex 3V6Z, for groundtruth in Figs. 6(a) and 7(a), we set the values as 1 if there exists an interaction between any two residues from the ligand and receptor proteins, otherwise 0. In contrast, the prediction by IIGRL model is shown in Fig. 6(b) and affinity scores are shown in Fig. 7(b). Overall, the prediction is in agreement with the true interface, and affinity scores could highlight the most interactive pairs.

VI. CONCLUSION

In this paper, we present an effective intra-inter graph representation learning for protein-protein binding sites prediction. For intra-graph learning, we propose to maximize the mutual information between the local node representation and

global graph summary, thus encouraging the global information encoded into node representation in the protein graph. For the inter-graph learning, we explore fusing the two separate ligand and receptor graphs as a whole graph, learning the affinity between residues/nodes of different proteins and propagating the information to each other, which could effectively capture inter-protein information and further enhance the discrimination of node pairs. Comprehensive experiments on multiple benchmarks are conducted for performance evaluation, demonstrating that the proposed IIGRL outperforms baselines.

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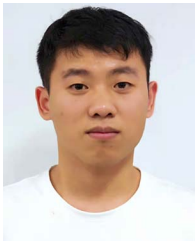
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